



DR ESLAM AHMED
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Transformations of Graphs

Outside the bracket changes y and does what it says. Inside the bracket changes x and does the opposite.

THE TOPIC ON ONE MAP

Transformations of Graphs

01 Translations and reflections

$f(x + 4), f(x) - 7$

$-f(x), f(-x)$

map a point

02 Stretches

$2f(x)$

$f(2x)$

$f\left(\frac{1}{2}x\right)$

03 Transform the equation

translate by $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

reflect in the y -axis

04 Trigonometric graphs

$y = a \cos(x + b)$

$\sin x - 2$

$\cos(x - 90)$

THE RULE Outside does what it says; inside does the opposite.

equation	what happens	$(3, 5)$ goes to
$f(x) + a$	up a	$(3, 5 + a)$
$f(x + a)$	left a (opposite)	$(3 - a, 5)$
$-f(x)$	reflect in the x -axis	$(3, -5)$
$f(-x)$	reflect in the y -axis	$(-3, 5)$
$k f(x)$	stretch vertically, $\times k$	$(3, 5k)$
$f(kx)$	stretch horizontally, $\div k$	$\left(\frac{3}{k}, 5\right)$

01 Translations and Reflections

Follow the turning point. A number outside the bracket moves y the way it says; a number inside moves x the opposite way. A minus sign reflects.

WHEN YOU SEE → YOU DO

$f(x) - 7$	→ y down 7: $(x, y - 7)$
$f(x + 4)$	→ left 4: $(x - 4, y)$
$-f(x)$	→ reflect in the x -axis: y changes sign
$f(-x)$	→ reflect in the y -axis: x changes sign
"describe fully the transformation"	→ name it + the vector or the mirror line
" $h(x + a)$ passes through $(2, 0)$ "	→ roots move: $-1 - a = 2$ or $6 - a = 2$

Outside → y as written

Inside → x the opposite

Minus → reflect

WORKED MODEL · NOVEMBER 2024 P2H

The minimum of $y = f(x)$ is $(5, 4)$. Find the minimum of (i) $f(x + 5)$ (ii) $3f(x)$ (iii) $f(x) - 7$.

1 inside +5 → x left 5: $(0, 4)$

opposite

2 outside $\times 3$ → y : $(5, 12)$

as written

ANSWER $(0, 4), (5, 12), (5, -3)$

× **TRAP** $f(x + 4)$ does **not** move right. Inside the bracket, the sign is the opposite.

IDEA 01

Practise: Translations and Reflections

My questions first, then selected practice, then real past papers.

Q1**DR ESLAM**

Move a point

CH 3 · TOPIC 9

3 marks

The minimum of $y = f(x)$ is $(2, -3)$. Find the minimum of (a) $f(x - 5)$ (b) $f(x) + 4$ (c) $f(-x)$.

SELECTED PRACTICE

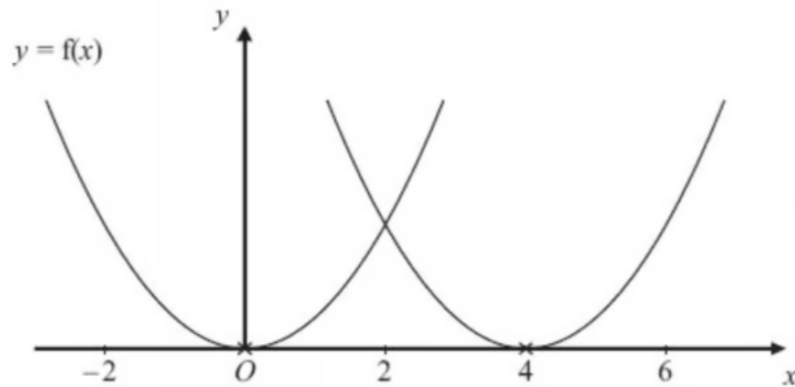
Q2

PRACTICE

Practice · Easy

CH 3 · TOPIC 9 2 marks

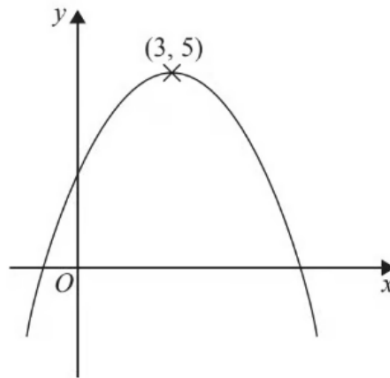
1



The curve with equation $y = f(x)$ is translated so that the point at $(0, 0)$ is mapped onto the point $(4, 0)$.

Find an equation of the translated curve.

(2 marks)

Q3**PRACTICE** Practice · Medium**CH 3 · TOPIC 9** 4 marks**1 (a)**

The diagram shows part of the curve with equation $y = f(x)$. The coordinates of the maximum point of the curve are $(3, 5)$.

Write down the coordinates of the maximum point of the curve with equation

(i) $y = f(x + 3)$

[1]

(ii) $y = -f(x)$

[1]

(iii) $y = f(-x)$

[1]

(3 marks)

- (b)** The curve with equation $y = f(x)$ is transformed to give the curve with equation $y = f(x) - 4$

Describe the transformation.

(1 mark)

Q4**PRACTICE**

Practice · Hard

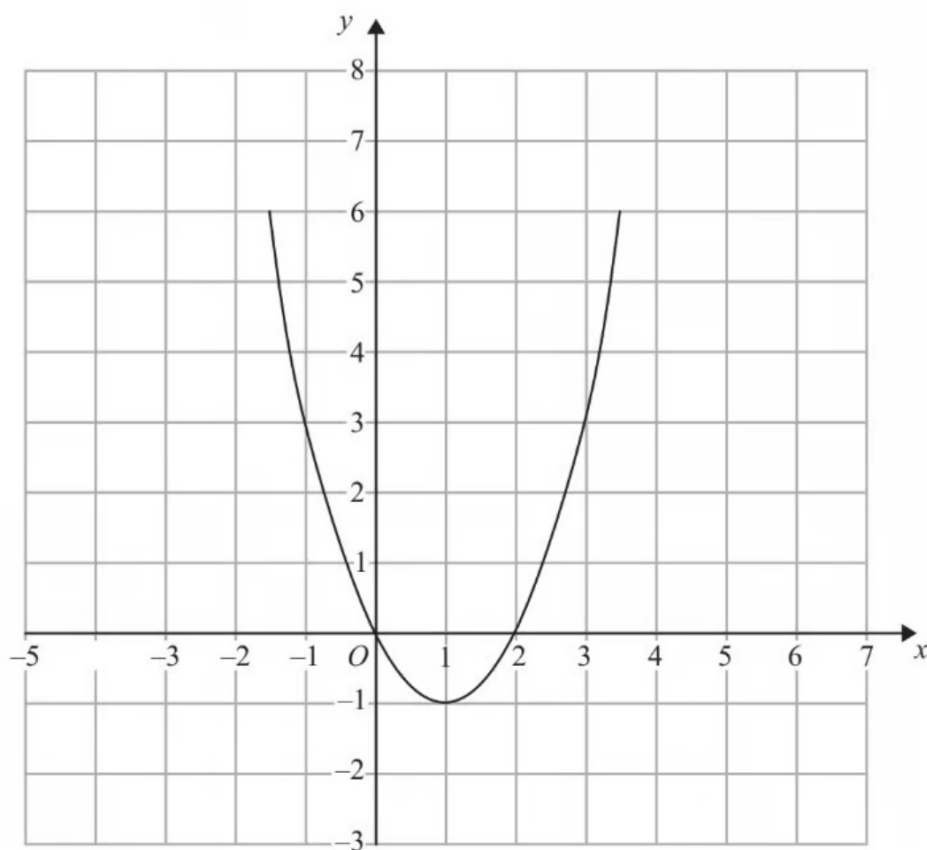
CH 3 · TOPIC 9 4 marks

1 (a) The graph of $y = f(x)$ is shown on each of the grids.

On this grid, sketch the graph of $y = f(x - 3)$

(2 marks)

(b) On this grid, sketch the graph of $y = f(-x) + 2$

**(2 marks)**

Q5**PRACTICE**

Practice · Hard

CH 3 · TOPIC 9 9 marks

11 (a) The function $f(x)$ is defined as $f(x) = 3 - 8x - 2x^2$

Find the coordinates of the turning point on the graph of $y = f(x)$

(3 marks)

(b) Using your result from part (a) write down the coordinates of the turning point on the following graphs:

(i) $y = f(x) + 2$

[1]

(ii) $y = f(x - 3)$

[1]

(iii) $y = f(4x)$

[1]

(iv) $y = -f(x)$

[1]

(v) $y = 3f(x + 1)$

[2]

(6 marks)

PAST PAPERS

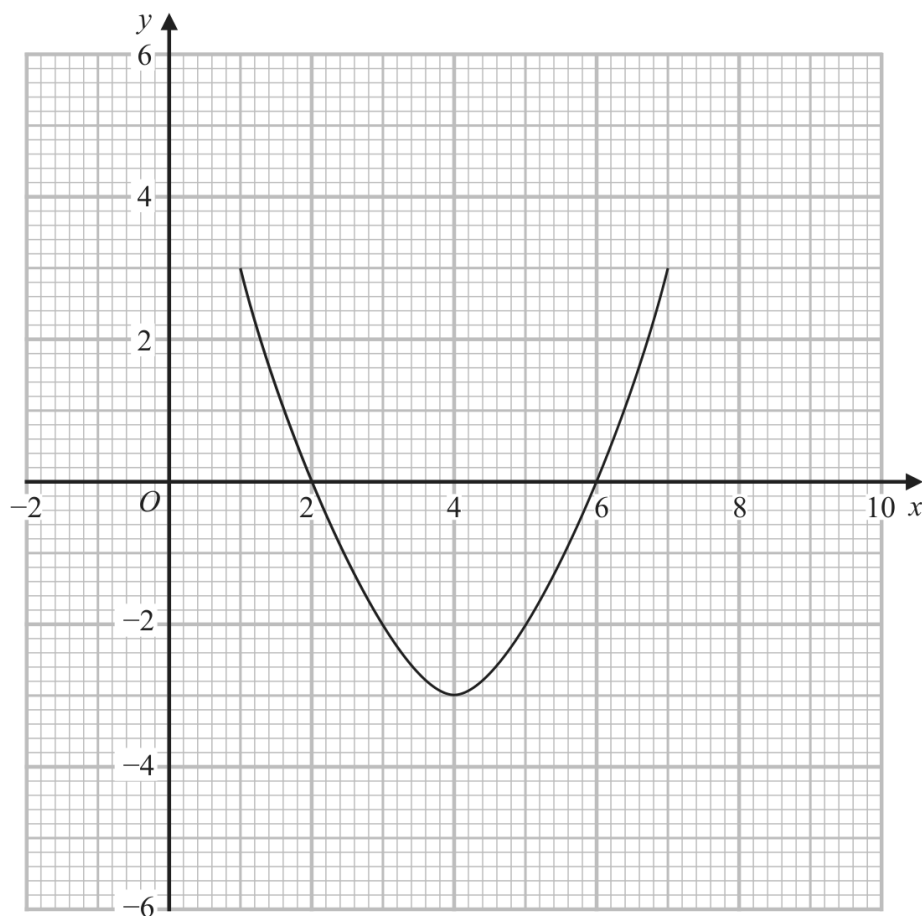
Q6 PAST PAPER Edexcel 4MA1 · Oct/Nov 2020 · Paper 1H · Q25

CH 3 · TOPIC 9 3 marks

- 25 The curve with equation $y = g(x)$ is transformed to the curve with equation $y = -g(x)$ by the single transformation **T**.

(a) Describe fully the transformation **T**.

The diagram shows the graph of $y = f(x)$



(b) On the grid, draw the graph of $y = 2f(x - 1)$

(Total for Question 25 is 3 marks)

Q7**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1H · Q21

CH 3 · TOPIC 9 4 marks**21** A curve has equation $y = f(x)$ The coordinates of the minimum point on this curve are $(-9, 15)$

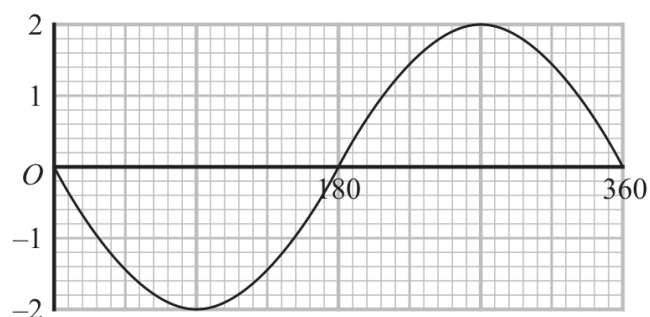
(a) Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 3)$

(.....,)

(ii) $y = \frac{1}{3}f(x)$

(.....,)

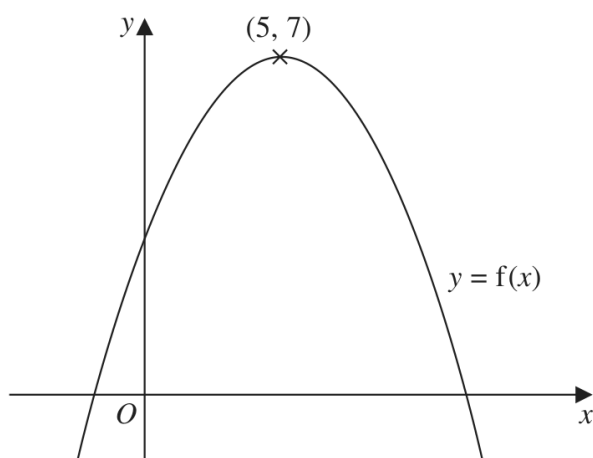
The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.Given that $a > 0$ and that $0 < b < 360$ (b) find the value of a and the value of b . $a = \dots\dots\dots$ $b = \dots\dots\dots$ **(Total for Question 21 is 4 marks)**

Q8**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q23

CH 3 · TOPIC 9 2 marks

23 The diagram shows a sketch of the curve with equation $y = f(x)$



There is only one maximum point on the curve.
The coordinates of this maximum point are (5, 7)

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 9)$

(.....,)

(ii) $y = f(x) + 3$

(.....,)

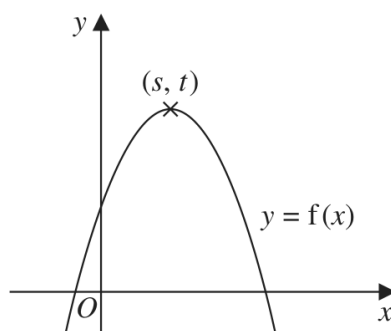
(Total for Question 23 is 2 marks)

Q9**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1H · Q20

CH 3 · TOPIC 9 2 marks

20 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (s, t)

Find, in terms of s and t , the coordinates of the maximum point on the curve with equation

(i) $y = f(x - 2)$

(.....,)

(ii) $y = 3f(x)$

(.....,)

(Total for Question 20 is 2 marks)

Q10**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 2HR · Q22

CH 3 · TOPIC 9 4 marks

22 The point A with coordinates $(-3, 2)$ lies on the straight line with equation $y = f(x)$

(a) Find the coordinates of the image of the point A on the straight line with equation

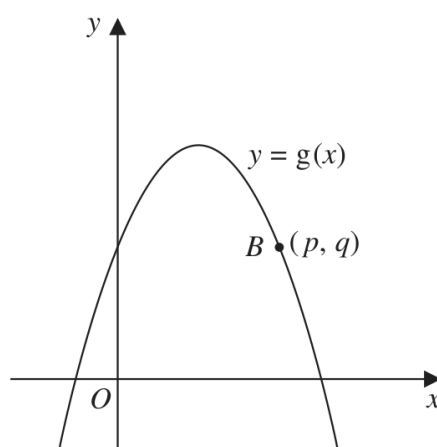
(i) $y = f(x) - 3$

(.....,)

(ii) $y = f\left(\frac{x}{2}\right)$

(.....,)

Here is a sketch of part of the curve with equation $y = g(x)$



The point B with coordinates (p, q) lies on the curve.

(b) Find the coordinates of the image of the point B on the curve with equation

$$y = -g(x - c)$$

where c is a constant.

(.....,)

(Total for Question 22 is 4 marks)

Q11**PAST PAPER**

Edexcel 4MA1 · June 2024 · Paper 2HR · Q23

CH 3 · TOPIC 9 2 marks**23** A curve has equation $y = f(x)$ The coordinates of the minimum point on this curve are $(6, -3)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 10$

(.....,)

(ii) $y = f(3x)$

(.....,)

(Total for Question 23 is 2 marks)**Q12****PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1H · Q21

CH 3 · TOPIC 9 2 marks**21** A curve has equation $y = f(x)$

There is one minimum point on this curve.

The coordinates of this minimum point are $(5, -4)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 7)$

(.....,)

(ii) $y = f(x) - 6$

(.....,)

(Total for Question 21 is 2 marks)

Q13**PAST PAPER**

Edexcel 4MA1 · November 2024 · Paper 2H · Q19

CH 3 · TOPIC 9 3 marks

19 A curve has equation $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are (5, 4)

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 5)$

(.....,)

(ii) $y = 3f(x)$

(.....,)

(iii) $y = f(x) - 7$

(.....,)

(Total for Question 19 is 3 marks)

Q14 PAST PAPER Edexcel 4MA1 · June 2025 · Paper 1H · Q19

CH 3 · TOPIC 9 2 marks

19 The equation of a curve is $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are $(8, -12)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 3$

(.....,)

(ii) $y = f(2x)$

(.....,)

(Total for Question 19 is 2 marks)

Q15**PAST PAPER**

Edexcel 4MA1 · November 2025 · Paper 1H · Q24

CH 3 · TOPIC 9 3 marks**24** A curve has equation $y = f(x)$

There is only one turning point on the curve.

The coordinates of this turning point are (4, 3)

Write down the coordinates of the turning point on the curve with equation

(i) $y = f(x + 5)$

(..... ,)

(ii) $y = f(x) + 7$

(..... ,)

(iii) $y = f(2x)$

(..... ,)

(Total for Question 24 is 3 marks)

02 Stretches

$k f(x)$ multiplies every y by k . $f(kx)$ divides every x by k . Points on the other axis stay where they are.

WHEN YOU SEE → YOU DO

$$2f(x) \longrightarrow y \times 2: (x, 2y)$$

$$f(2x) \longrightarrow x \div 2: \left(\frac{x}{2}, y\right)$$

$$f\left(\frac{1}{2}x\right) \longrightarrow x \times 2$$

$$\text{"sketch } y = 2g(x)\text{"} \longrightarrow \text{double every } y; x\text{-intercepts stay}$$

WORKED MODEL · MAY 2023 P1HR

The maximum of $y = f(x)$ is $(3, 5)$. Find the maximum of $y = 2f(x)$ and of $y = f(2x)$.

1 $2f(x): y \times 2 \rightarrow (3, 10)$

outside

2 $f(2x): x \div 2 \rightarrow (1.5, 5)$

inside: opposite

ANSWER $(3, 10)$ and $(1.5, 5)$

IDEA 02

Practise: Stretches

My questions first, then selected practice, then real past papers.

Q16 DR ESLAM Stretch

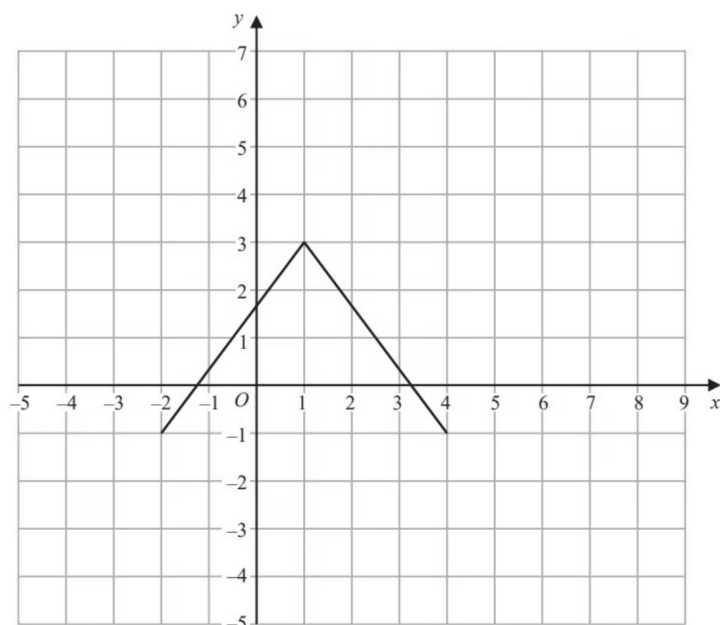
CH 3 · TOPIC 9 2 marks

The maximum of $y = f(x)$ is $(6, 4)$. Find the maximum of $y = f(3x)$ and of $y = \frac{1}{2}f(x)$.

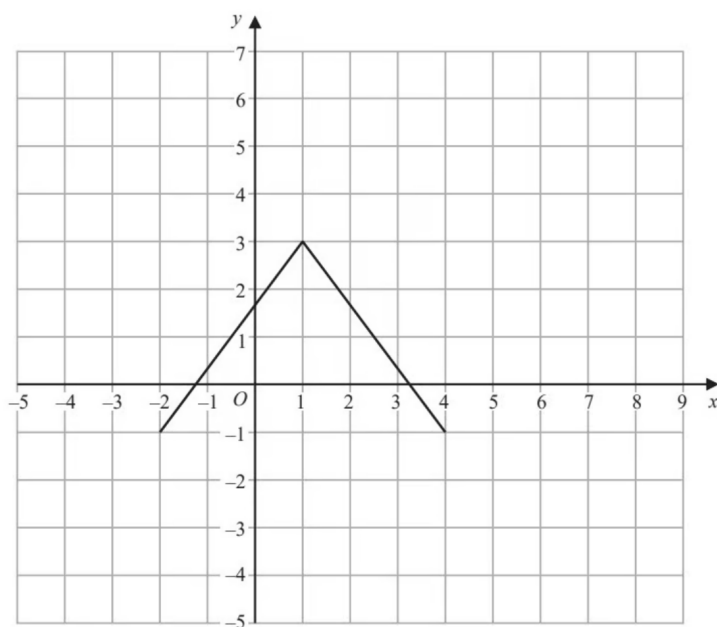
SELECTED PRACTICE

Q17 PRACTICE Practice · Easy

CH 3 · TOPIC 9 4 marks

6 (a) Here is the graph of $y = f(x)$ On the grid above, draw the graph of $y = 2f(x)$

(2 marks)

(b) Here is the graph of $y = f(x)$ On the grid above, draw the graph of $y = f(-x)$

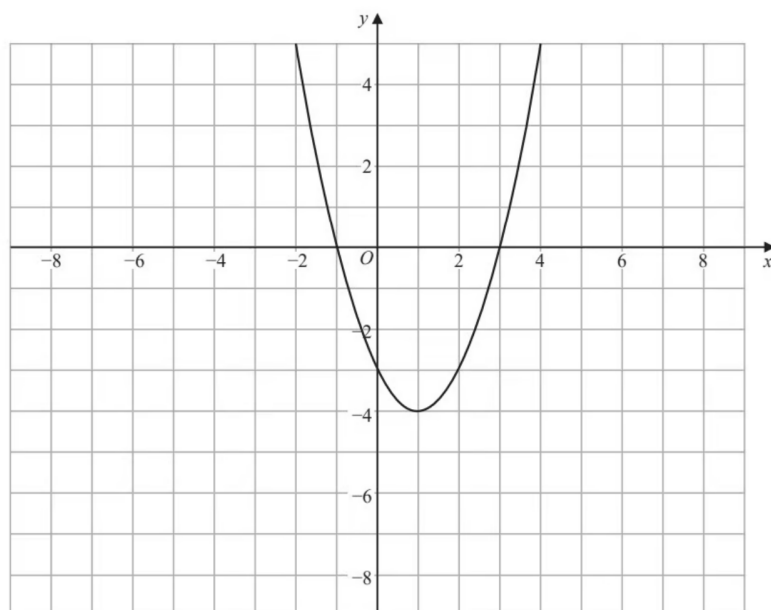
(2 marks)

Q18**PRACTICE**

Practice · Medium

CH 3 · TOPIC 9 3 marks

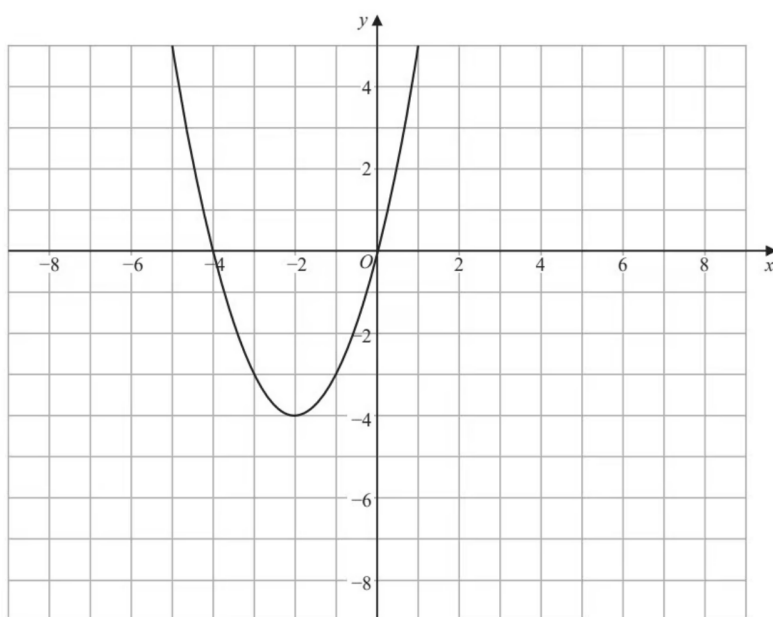
9 (a) The graph of $y = f(x)$ is shown on the grid.



On the grid above, sketch the graph of $y = f\left(\frac{1}{2}x\right)$

(2 marks)

(b) The graph of $y = f(x + k)$ is shown on the grid below.



Write down the value of k

(1 mark)

PAST PAPERS

Q19**PAST PAPER**

Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q20

CH 3 · TOPIC 9 2 marks**20** A curve has equation $y = f(x)$

There is only one maximum point on the curve.

The coordinates of this maximum point are $(-3, 4)$

Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x) - 6$

(..... ,)

(ii) $y = f(2x)$

(..... ,)

(Total for Question 20 is 2 marks)

Q20**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1HR · Q20

CH 3 · TOPIC 9 6 marks

20 The curve with equation $y = f(x)$ has one turning point.

The coordinates of this turning point are $(-6, -4)$

(a) Write down the coordinates of the turning point on the curve with equation

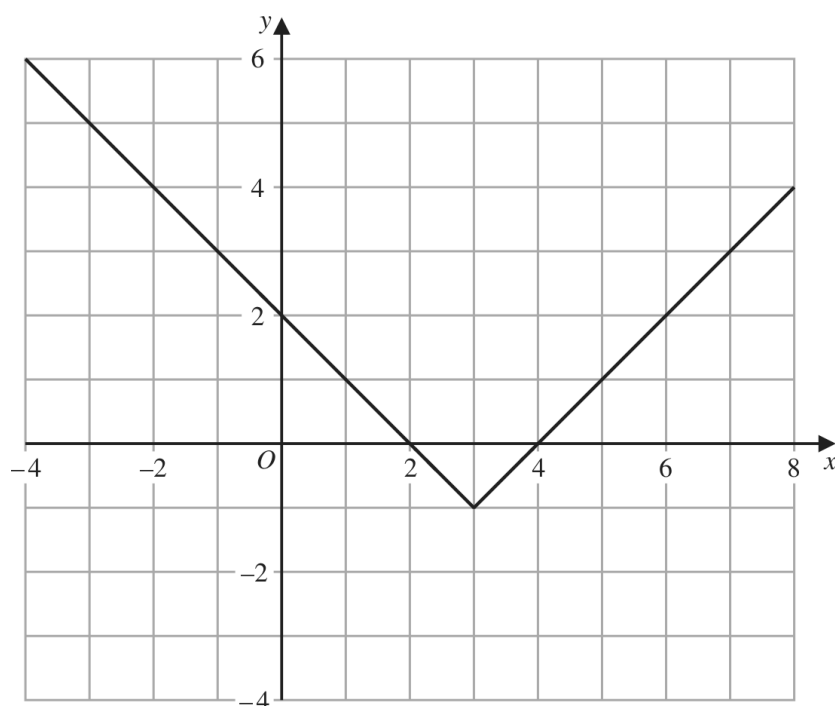
(i) $y = f(x) + 5$

(.....,))

(ii) $y = f(3x)$

(.....,))

The graph of $y = g(x)$ is shown on the grid below.



(b) On the grid, sketch the graph of $y = 2g(x)$ for $-1 \leq x \leq 7$

The graph of $y = h(x)$ intersects the x -axis at two points.

The coordinates of the two points are $(-1, 0)$ and $(6, 0)$

The graph of $y = h(x + a)$ passes through the point with coordinates $(2, 0)$, where a is a constant.

(c) Find the two possible values of a

.....,

(Total for Question 20 is 6 marks)

Q21 **PAST PAPER** Edexcel 4MA1 · May 2023 · Paper 1H · Q21**CH 3 · TOPIC 9** **2 marks**

21 A curve has equation $y = f(x)$

There is only one turning point on the curve.

The coordinates of this turning point are (6, 5)

Write down the coordinates of the turning point on the curve with equation

(a) $y = f(x - 4)$

(.....,)

(b) $y = f(3x)$

(.....,)

(Total for Question 21 is 2 marks)

Q22**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 1HR · Q21

CH 3 · TOPIC 9 3 marks

- 21 The curve with equation $y = f(x)$ has one turning point.
The coordinates of this turning point are $(-6, 9)$

- (a) Write down the coordinates of the turning point on the curve
with equation $y = f(3x)$

(.....,)

The curve **C** with equation $y = g(x)$ is transformed to give the curve **S** with
equation $y = g(x + a) + b$

The point $(4, -5)$ on **C** is mapped to the point $(1, -16)$ on **S**

- (b) Write down the value of a and the value of b

$a = \dots\dots\dots$

$b = \dots\dots\dots$

(Total for Question 21 is 3 marks)

03 Transform the Equation

Translate by $\begin{pmatrix} a \\ b \end{pmatrix}$: replace x with $(x-a)$ and add b . Reflect in the y -axis: replace x with $-x$. Then expand if asked.

WHEN YOU SEE → YOU DO

“translation $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$ — find an equation for S ”

→ replace x by $(x - 4)$

“translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$ ”

→ replace x by $(x - 4)$, add 6

“reflection in the y -axis”

→ replace x by $-x$

“ $g(x) = 5(x + 4)^2 + 9(x + 4) + 8$ from $f(x)$ ”

→ inside $+4 \rightarrow a = -4$; $+6$ outside $\rightarrow b = 6$

“describe a series of transformations”

→ complete the square, then read each change

WORKED MODEL · MAY 2023 P1HR

$L: y = x^2 + 7x + 20$ is translated by $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$. Find S as $y = ax^2 + bx + c$.

1 $y = (x - 2)^2 + 7(x - 2) + 20$

replace x by $x - 2$

2 $y = x^2 - 4x + 4 + 7x - 14 + 20$

expand

ANSWER $y = x^2 + 3x + 10$

IDEA 03

Practise: Transform the Equation

My questions first, then selected practice, then real past papers.

Q23 DR ESLAM Translate the equation

CH 3 · TOPIC 9 3 marks

$y = x^2 + 2x$ is translated by $\begin{pmatrix} 3 \\ 1 \end{pmatrix}$. Find the new equation in the form $y = ax^2 + bx + c$.

SELECTED PRACTICE**Q24** PRACTICE Practice · Hard

CH 3 · TOPIC 9 4 marks

8 The curve **S** has equation $y = f(x)$ where $f(x) = x^2$

The curve **T** has equation $y = g(x)$ where $g(x) = 2x^2 - 12x + 13$

By writing $g(x)$ in the form $a(x - b)^2 - c$, where a , b and c are constants, describe fully a series of transformations that map the curve **S** onto the curve **T**.

(4 marks)

Q25**PRACTICE**

Practice · Hard

CH 3 · TOPIC 9 4 marks

12 The graph of $y = x^3 + 6$ is translated 4 units to the right.

The translated graph has equation $y = f(x)$

Work out $f(x)$.

Give your answer in the form $x^3 + ax^2 + bx + c$ where a , b and c are integers.

(4 marks)**Q26****PRACTICE**

Practice · Hard

CH 3 · TOPIC 9 3 marks

13 Curve P has equation $y = 2(x - 1)^2 - 5$

Curve Q is a reflection in the y -axis of curve P.

Work out the equation of curve Q.

Give your answer in the form $y = ax^2 + bx + c$ where a , b and c are integers.

(3 marks)

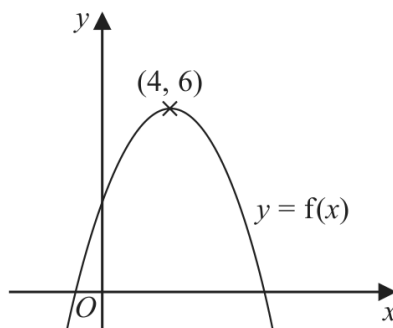
PAST PAPERS

Q27**PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2H · Q21

CH 3 · TOPIC 9 4 marks

21 The diagram shows a sketch of part of the curve with equation $y = f(x)$



There is one maximum point on this curve.

The coordinates of this maximum point are (4, 6)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = f(x + 4)$

(.....,)

(ii) $y = f(2x)$

(.....,)

The equation of a curve **C** is $y = x^2 + 3x + 4$

The curve **C** is transformed to curve **S** under the translation $\begin{pmatrix} 4 \\ 6 \end{pmatrix}$

(b) Find an equation of curve **S**.

You do not need to simplify the equation.

(Total for Question 21 is 4 marks)

Q28**PAST PAPER**

Edexcel 4MA1 · May 2023 · Paper 1HR · Q20

CH 3 · TOPIC 9 7 marks**20** Curve **C** has equation $y = f(x)$ The graph of curve **C** has one maximum point.

The coordinates of this maximum point are (3, 5)

(a) Write down the coordinates of the maximum point on the curve with equation

(i) $y = 2f(x)$

(.....,)

(ii) $y = f(x) - 7$

(.....,)

(iii) $y = f(-x)$

(.....,)

Curve **L** has equation $y = x^2 + 7x + 20$ Curve **L** is transformed to curve **S** under the translation $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ (b) Find an equation for **S**Give your answer in the form $y = ax^2 + bx + c$ $y = \dots\dots\dots$ **(Total for Question 20 is 7 marks)**

04 Trigonometric Graphs

The same rules apply. $\sin x - 2$ moves down 2; $\cos(x - 90^\circ)$ moves right 90° . For $y = a \cos(x + b)$: a = the height of a peak, b = how far left the graph moved.

WHEN YOU SEE → YOU DO

$$y = \sin x + 2, \sin x - 2$$

→ up / down 2

$$y = \cos(x - 30)$$

→ right 30° : crosses at 120, 300

$$y = -\cos x$$

→ reflect in the x-axis

“find a and b ” in $y = a \cos(x + b)$

→ a = max; the max moved from $0^\circ \rightarrow b$

WORKED MODEL · JANUARY 2020 P1HR

$y = a \cos(x + b)^\circ$ has maximum 2.5 at $x = 60$ (for $0 \leq x \leq 360$). Find a and b .

1 $a = 2.5$

height of the peak

2 $\cos x$ has its max at 0° (and 360°); it moved to 60°

right 60°

ANSWER $a = 2.5, b = -60$ (or $b = 300$)

IDEA 04

Practise: Trigonometric Graphs

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE

Q29

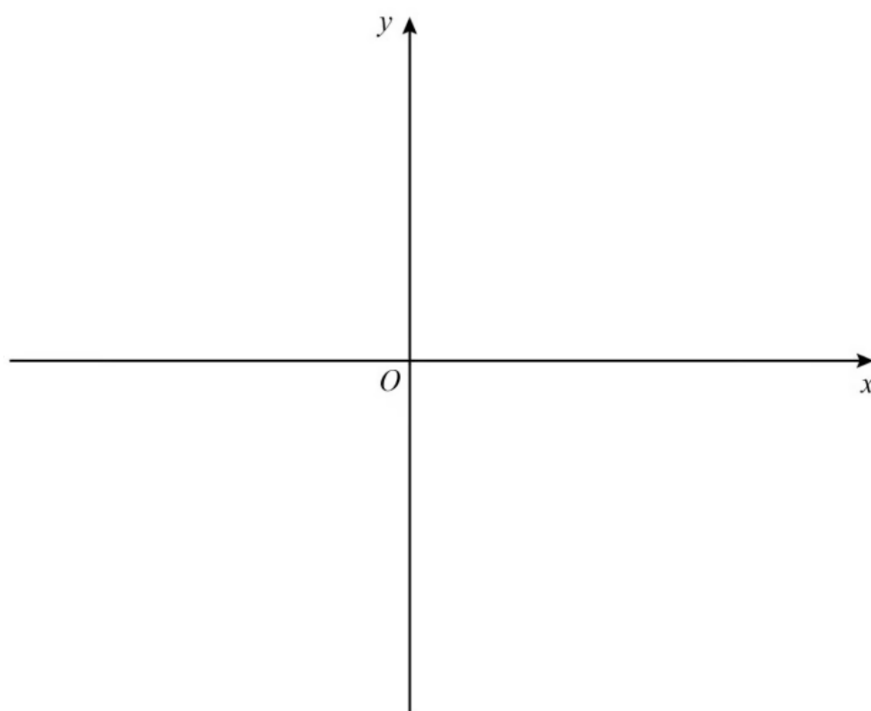
PRACTICE

Practice · Easy

CH 3 · TOPIC 9 2 marks

- 11 On the axes, sketch the curve $y = x^3 - 2$

You must show the coordinates of the y -intercept.



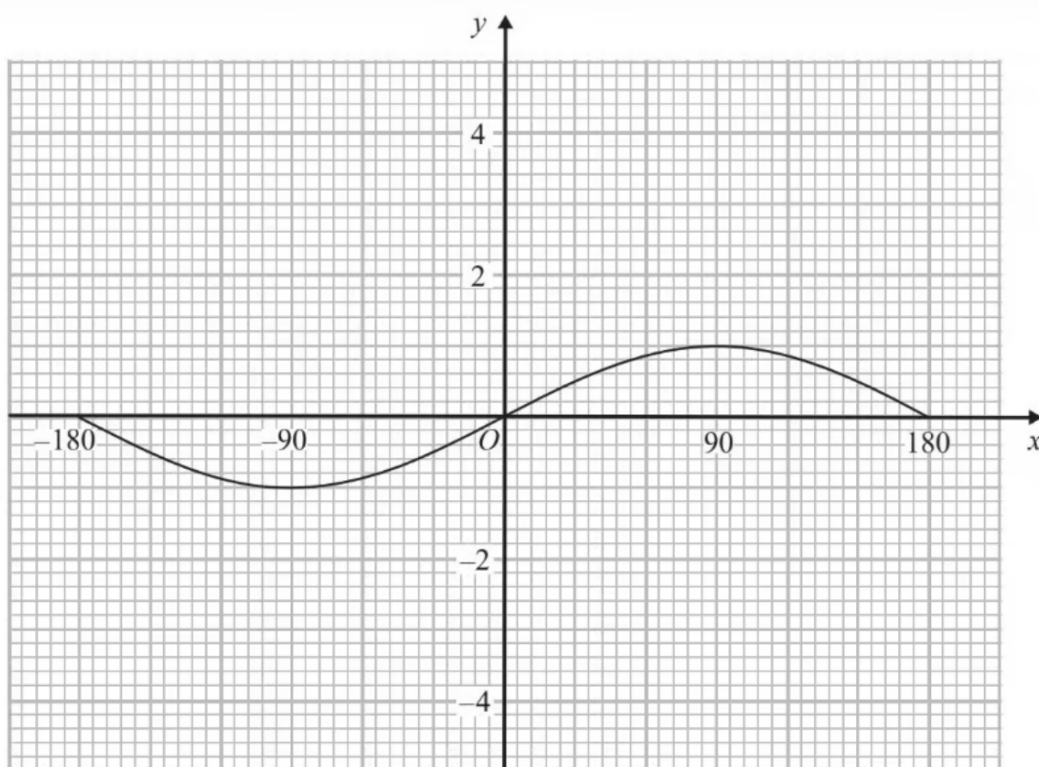
(2 marks)

Q30**PRACTICE**

Practice · Medium

CH 3 · TOPIC 9 2 marks

3 Here is the graph of $y = \sin x^\circ$ for $-180 \leq x \leq 180$



On the grid, sketch the graph of $y = \sin x^\circ - 2$ for $-180 \leq x \leq 180$

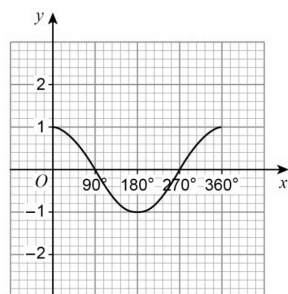
(2 marks)

Q31

PRACTICE Practice · Medium

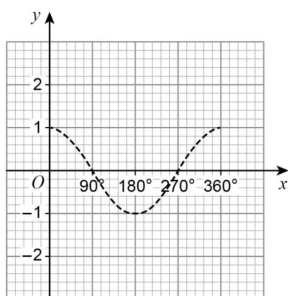
CH 3 · TOPIC 9 3 marks

11 (a) Here is the graph of $y = \cos x$ for $0^\circ \leq x \leq 360^\circ$



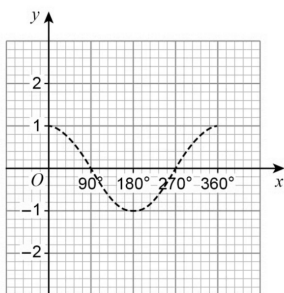
In parts (a) and (b) the graph of $y = \cos x$ is shown as a dashed line.

On the grid below, draw the graph of $y = \cos(x - 90^\circ)$ for $0^\circ \leq x \leq 360^\circ$



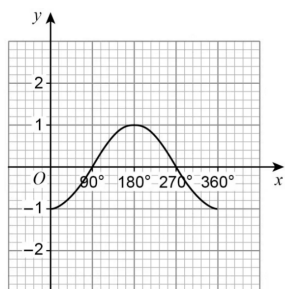
(1 mark)

(b) On the grid below, draw the graph of $y = 1 + \cos x$ for $0^\circ \leq x \leq 360^\circ$



(1 mark)

(c) Rita tries to draw the graph of $y = \cos(-x)$ for $0^\circ \leq x \leq 360^\circ$. Here is her graph.



Give a reason why Rita's graph is incorrect.

(1 mark)

Q32 PRACTICE Practice · Hard

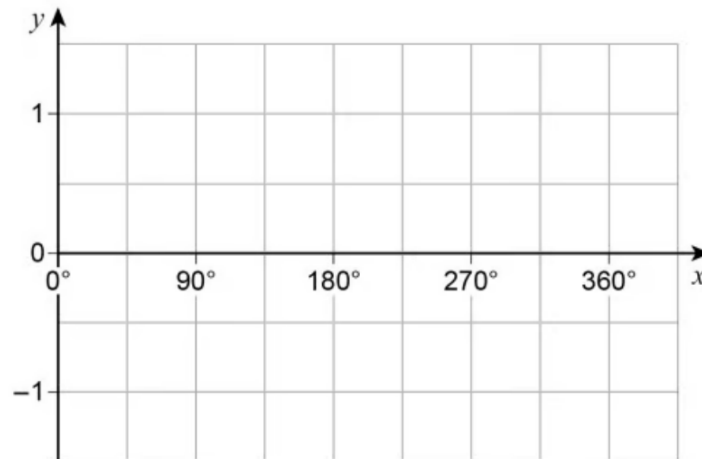
CH 3 · TOPIC 9 2 marks

14 For all values of x

$$f(x) = \sin x$$

$$g(x) = x + 90$$

On the grid, draw the graph of the composite function $y = fg(x)$ for $0^\circ \leq x \leq 360^\circ$



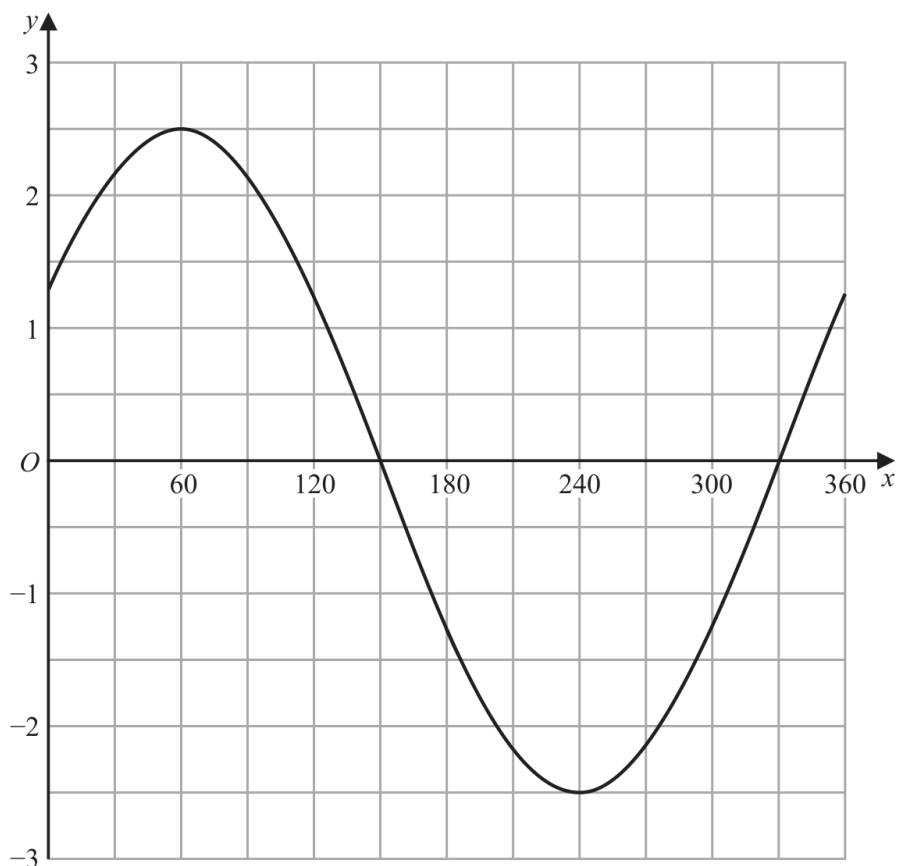
(2 marks)

PAST PAPERS

Q33 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 1HR · Q22

CH 3 · TOPIC 9 4 marks

22 The graph of $y = a \cos(x + b)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid.



(a) Find the value of a and the value of b .

$a = \dots\dots\dots$

$b = \dots\dots\dots$

Another curve C has equation $y = f(x)$

The coordinates of the minimum point of C are $(4, 5)$

(b) Write down the coordinates of the minimum point of the curve with equation

(i) $y = f(2x)$

(.....,)

(ii) $y = f(x) - 7$

(.....,)

(Total for Question 22 is 4 marks)

Q34 PAST PAPER Edexcel 4MA1 · November 2021 · Paper 2H · Q21

CH 3 · TOPIC 9 6 marks

- 21 The curve **C** has equation $y = f(x)$ where $f(x) = 9 - 3(x + 2)^2$
The point **A** is the maximum point on **C**.

(a) Write down the coordinates of **A**.

(.....,)
(1)

The curve **C** is transformed to the curve **S** by a translation of $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

(b) Find an equation for the curve **S**.

.....

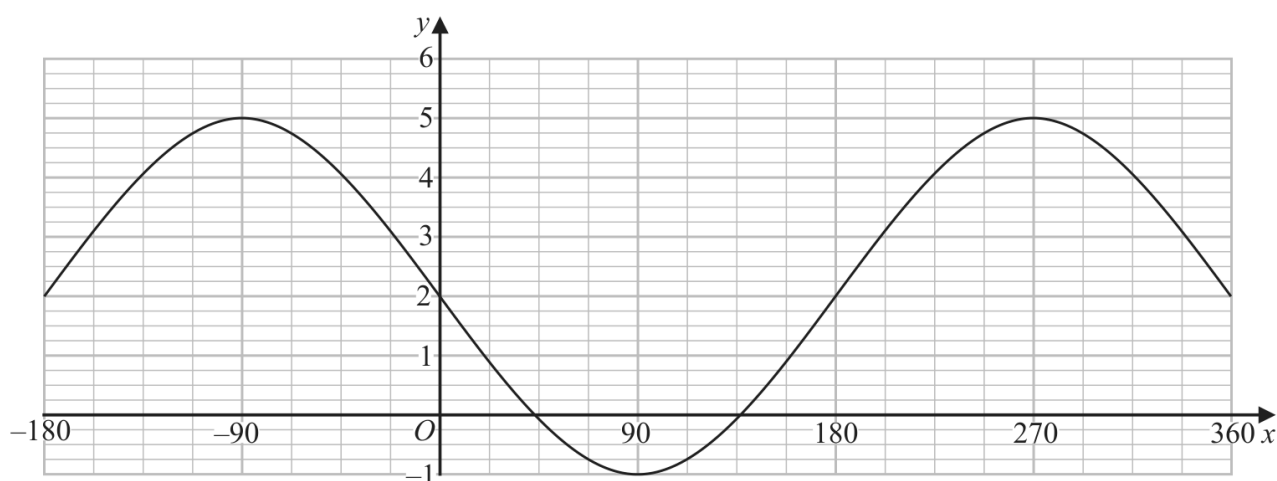
The curve **C** is transformed to the curve **T**.

The curve **T** has equation $y = 3(x + 2)^2 - 9$

(c) Describe fully the transformation that maps curve **C** onto curve **T**.

.....

The graph of $y = a \cos(x - b)^\circ + c$ for $-180 \leq x \leq 360$ is drawn on the grid below.



(d) Find the value of a , the value of b and the value of c .

$a =$

$b =$

$c =$

(Total for Question 21 is 6 marks)

Q35**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 1H · Q24

CH 3 · TOPIC 9 4 marks

24 The curve with equation $f(x) = 5x^2 + 9x + 2$ is transformed to the curve with equation

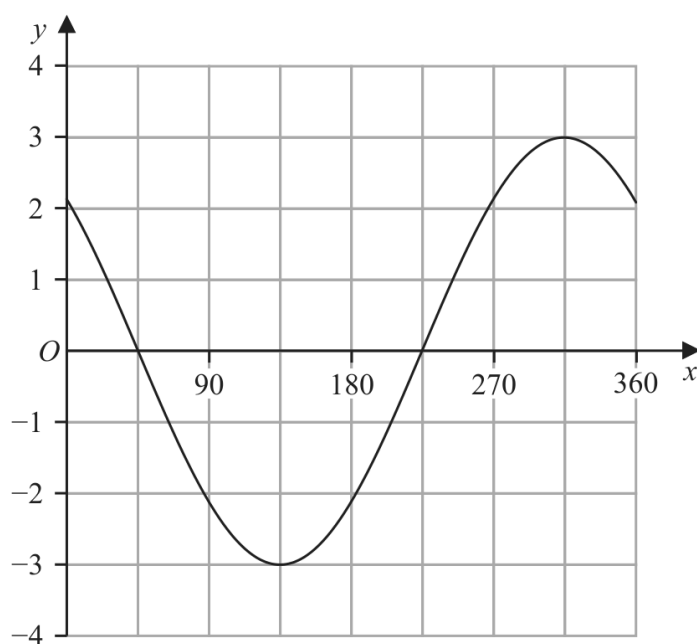
$$g(x) = 5(x+4)^2 + 9(x+4) + 8 \text{ by the translation } \begin{pmatrix} a \\ b \end{pmatrix}$$

(a) Write down the value of a and the value of b

$a = \dots\dots\dots$

$b = \dots\dots\dots$

The graph of $y = p \cos(x+q)^\circ$ for $0 \leq x \leq 360$ is drawn on the grid below.



Given that $p > 0$ and $0 < q < 360$

(b) find the value of p and the value of q

$p = \dots\dots\dots$

$q = \dots\dots\dots$

(Total for Question 24 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|--|
| 01 | Move / reflect | Outside: y as written. Inside: x opposite. Minus reflects. |
| 02 | Stretch | $k f(x)$: $y \times k$. $f(kx)$: $x \div k$. |
| 03 | Equation | $\begin{pmatrix} a \\ b \end{pmatrix}$: $x \rightarrow x - a$, then $+b$. y -axis: $x \rightarrow -x$. |
| 04 | Trig | Same rules; a = peak height, b = shift. |

BOTTOM LINE Outside does what it says → inside does the opposite → follow the turning point.

Answers

Transformations of Graphs · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · TRANSLATIONS AND REFLECTIONS

- Q1** Dr Eslam · Move a point (a) (7, -3) (b) (2, 1) (c) (-2, -3)
- Q2** Practice · Easy $y = f(x - 4)$
- Q3** Practice · Medium (a)(i) (0, 5) (ii) (3, -5) (iii) (-3, 5) (b) translation $\begin{pmatrix} 0 \\ -4 \end{pmatrix}$
- Q4** Practice · Hard (a) curve moved right 3 (b) reflected in y-axis, then up 2
- Q5** Practice · Hard (a) (-2, 11) (b)(i) (-2, 13) (ii) (1, 11) (iii) (-0.5, 11) (iv) (-2, -11) (v) (-3, 33)
- Q6** Edexcel 4MA1 · Oct/Nov 2020 · Paper 1H · Q25 (a) reflection in the x-axis (b) min moves to (5, -6)
- Q7** Edexcel 4MA1 · January 2021 · Paper 1H · Q21 (a)(i) (-12, 15) (ii) (-9, 5) (b) $a = 2$, $b = 90$
- Q8** Edexcel 4MA1 · January 2022 · Paper 1H · Q23 (i) (-4, 7) (ii) (5, 10)
- Q9** Edexcel 4MA1 · June 2022 · Paper 1H · Q20 (i) $(s + 2, t)$ (ii) $(s, 3t)$
- Q10** Edexcel 4MA1 · June 2022 · Paper 2HR · Q22 (a)(i) (-3, -1) (ii) (-6, 2) (b) $(p + c, -q)$
- Q11** Edexcel 4MA1 · June 2024 · Paper 2HR · Q23 (i) (6, 7) (ii) (2, -3)
- Q12** Edexcel 4MA1 · May 2024 · Paper 1H · Q21 (i) (-2, -4) (ii) (5, -10)
- Q13** Edexcel 4MA1 · November 2024 · Paper 2H · Q19 (i) (0, 4) (ii) (5, 12) (iii) (5, -3)
- Q14** Edexcel 4MA1 · June 2025 · Paper 1H · Q19 (i) (8, -9) (ii) (4, -12)
- Q15** Edexcel 4MA1 · November 2025 · Paper 1H · Q24 (i) (-1, 3) (ii) (4, 10) (iii) (2, 3)

IDEA 02 · STRETCHES

- Q16** Dr Eslam · Stretch (2, 4) and (6, 2)
- Q17** Practice · Easy (a) peak (1, 6), ends at (-2, 0) and (4, 0) (b) reflection in the y-axis
- Q18** Practice · Medium (a) $f(x)$ stretched horizontally $\times 2$: min (2, -4) (b) $k = 3$
- Q19** Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q20 (i) (-3, -2) (ii) (-1.5, 4)
- Q20** Edexcel 4MA1 · January 2022 · Paper 1HR · Q20 (a)(i) (-6, 1) (ii) (-2, -4) (c) $a = 3$ or -4
- Q21** Edexcel 4MA1 · May 2023 · Paper 1H · Q21 (a) (10, 5) (b) (2, 5)
- Q22** Edexcel 4MA1 · June 2025 · Paper 1HR · Q21 (a) (-2, 9) (b) $a = 3$, $b = -11$

IDEA 03 · TRANSFORM THE EQUATION

- Q23** Dr Eslam · Translate the equation $(x - 3)^2 + 2(x - 3) + 1 = x^2 - 4x + 4$
- Q24** Practice · Hard $g(x) = 2(x - 3)^2 - 5$: translate $\begin{pmatrix} 3 \\ 0 \end{pmatrix}$, stretch $\times 2$ vertically, translate $\begin{pmatrix} 0 \\ -5 \end{pmatrix}$
- Q25** Practice · Hard $x^3 - 12x^2 + 48x - 58$
- Q26** Practice · Hard $y = 2x^2 + 4x - 3$
- Q27** Edexcel 4MA1 · January 2020 · Paper 2H · Q21 (a)(i) (0, 6) (ii) (2, 6) (b) $y = (x - 4)^2 + 3(x - 4) + 10$
- Q28** Edexcel 4MA1 · May 2023 · Paper 1HR · Q20 (a)(i) (3, 10) (ii) (3, -2) (iii) (-3, 5) (b) $y = x^2 + 3x + 10$

IDEA 04 · TRIGONOMETRIC GRAPHS

- Q29** Practice · Easy cubic through $(0, -2)$
- Q30** Practice · Medium $\sin x$ moved down 2 (between -3 and -1)
- Q31** Practice · Medium (a) $\cos x$ moved right 90° (same as $\sin x$) (b) moved up 1 (c) $\cos(-x) = \cos x$, so it should start at $(0, 1)$
- Q32** Practice · Hard $\sin(x + 90) = \cos x$: from $(0, 1)$ down to $(180, -1)$ and back to $(360, 1)$
- Q33** Edexcel 4MA1 · January 2020 · Paper 1HR · Q22 (a) $a = 2.5, b = -60$ (b)(i) $(2, 5)$ (ii) $(4, -2)$
- Q34** Edexcel 4MA1 · November 2021 · Paper 2H · Q21 (a) $A(-2, 9)$ (b) $y = 9 - 3(x - 2)^2$ (c) reflection in the x -axis (d) $a = 3, b = 90, c = 2$
- Q35** Edexcel 4MA1 · November 2023 · Paper 1H · Q24 (a) $a = -4, b = 6$ (b) $p = 3, q = 45$